AXI-Stream

Проектирование цифровых вычислительных систем

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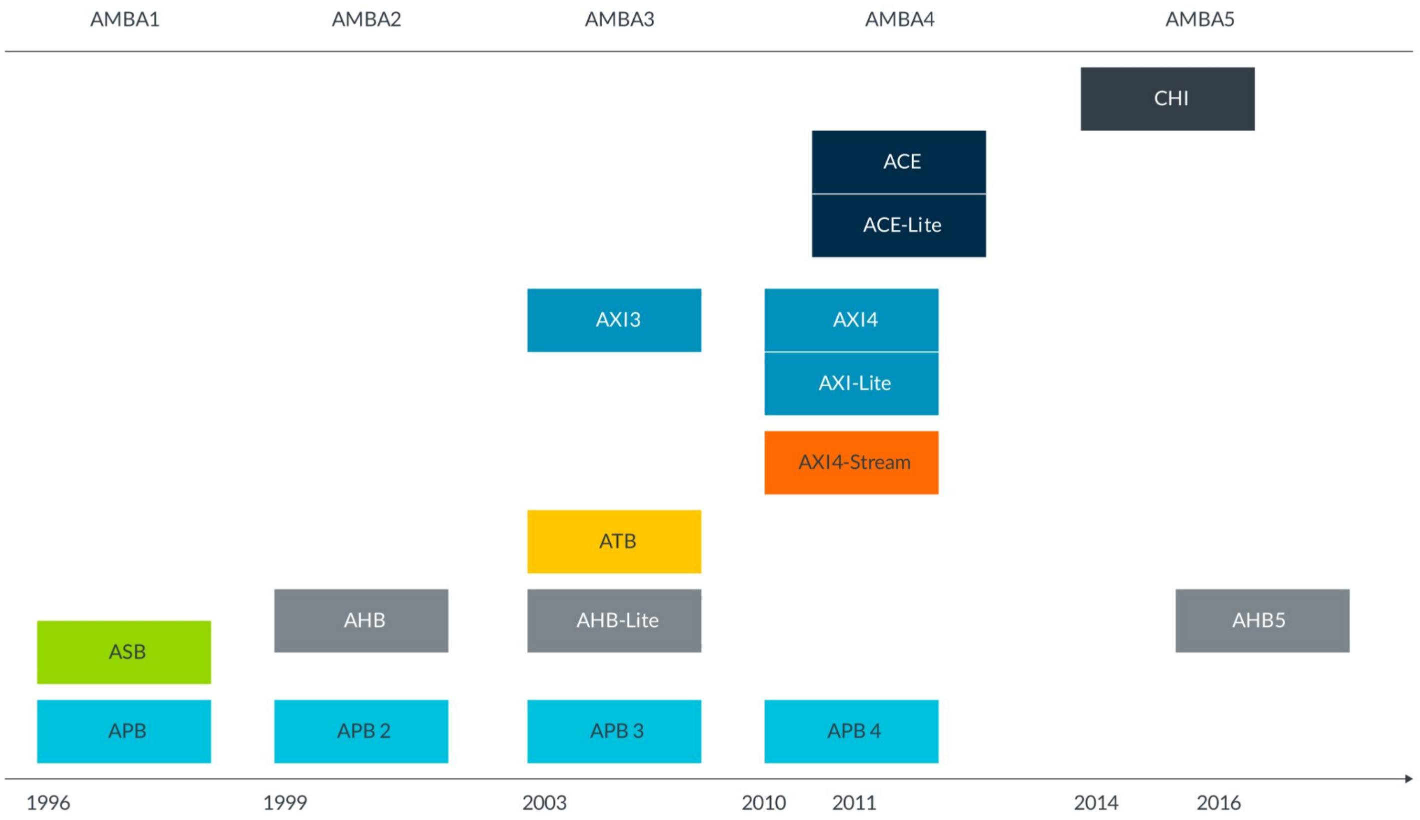
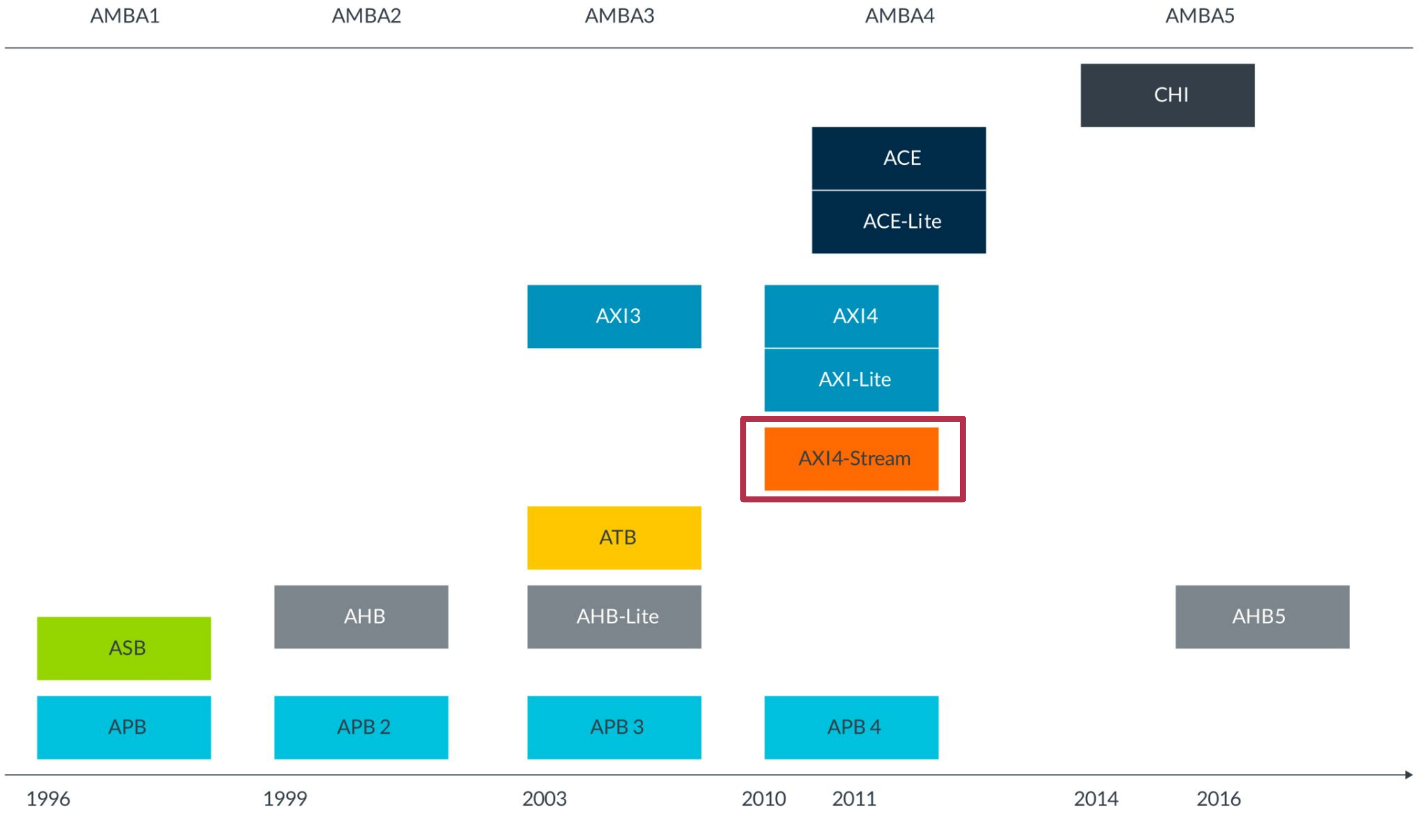
AMBA



Источник: Introduction to AMBA AXI4

AMBA: AXI-Stream



Источник: Introduction to AMBA AXI4

AXI-Stream

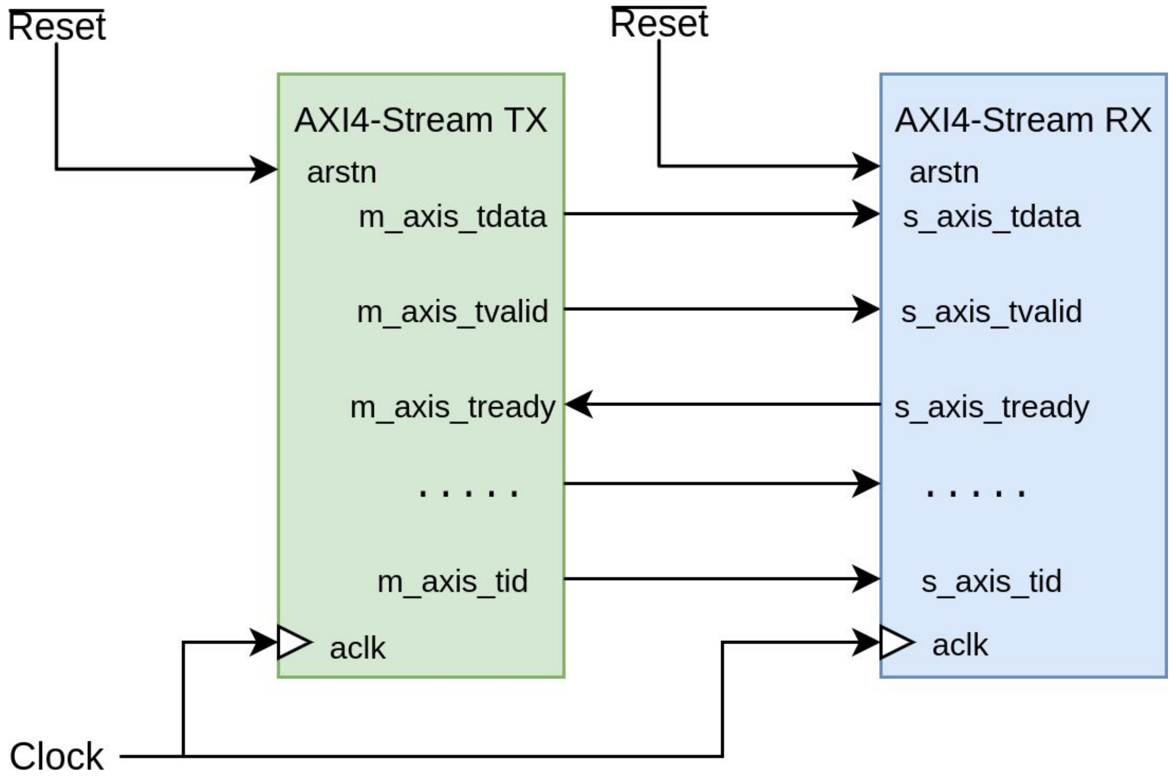
AXI-Stream:

- **AXI4-Stream**
- AXI5-Stream
  - Wake-up signaling
  - Interface protection using parity

Основные сигналы

Сигнал	Источник	Ширина	Примечание
ACLK	Clock	1	Глобальный clock
ARESETn	Reset	1	Глобальный reset
TVALID	TX	1	Handshake
TREADY	RX	1	
TDATA	TX	TDATA_WIDTH	Целое число байт
TID	TX	TID_WIDTH	Идентификатор
TSTRB	TX	TDATA_WIDTH/8	Data or null byte
TKEEP	TX	TDATA_WIDTH/8	Transmit byte to dest.
TLAST	TX	1	Конец пакета
TDEST	TX	TDEST_WIDTH	Routing information
TUSER	TX	TUSER_WIDTH	User information

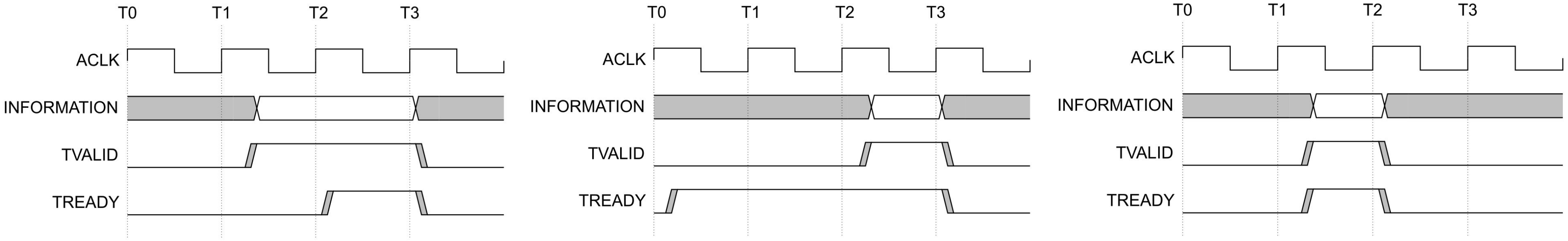
Соединение



Reset
Reset
AXI4-Stream TX
arstn
m_axis_tdata
m_axis_tvalid
m_axis_tready
. . . . .
m_axis_tid
aclk
AXI4-Stream RX
arstn
s_axis_tdata
s_axis_tvalid
s_axis_tready
. . . . .
s_axis_tid
aclk
Clock

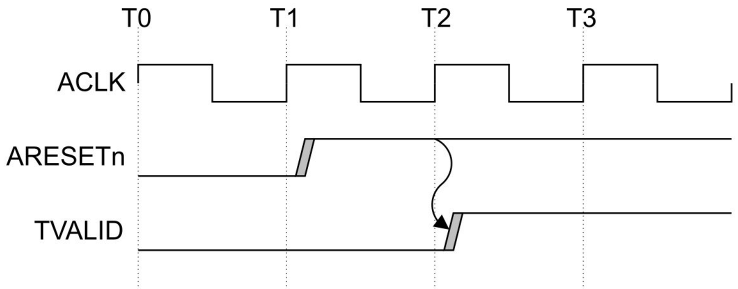
Handshake signaling

- For a **transfer** to occur, both `TVALID` and `TREADY` **must** be asserted. Either `TVALID` or `TREADY` **can** be asserted first, or both **can** be asserted in the same `ACLK` cycle.
- TX **is not permitted** to wait until `TREADY` is asserted before asserting `TVALID`. Once `TVALID` is asserted, it **must** remain asserted until the handshake occur.
- RX **is permitted** to wait for `TVALID` to be asserted before asserting `TREADY`. It **is permitted** that RX asserts and deasserts `TREADY` without `TVALID` being asserted.



Clock and Reset

- A single clock signal, `ACLK`, is used by each component. All input signals are sampled on the rising edge of `ACLK`. All output signal changes **must** occur after the rising edge of `ACLK`.
- `ARESETn` is a single, active-`LOW` reset signal. The reset signal **can** be asserted asynchronously, but deassertion **must** be synchronous after the rising edge of `ACLK`.
- During reset, `TVALID` **must** be driven `LOW`. All other signals **can** be driven to any value.
- TX interface **must** only begin driving `TVALID` at a rising `ACLK` edge following a rising edge at which `ARESETn` is asserted `HIGH`.



Термины

- Transfer
  - A single transfer of data across an AXI-Stream interface.
  - A single transfer is defined by a single TVALID and TREADY signal handshake.
- Packet
  - A group of bytes that are transported together across an AXI-Stream interface.
  - A packet can consist of a single transfer or multiple transfers.
- Frame
  - A frame contains an integer number of packets.
  - A frame can have a very large number of bytes, for example an entire video frame buffer.
  - A frame is the highest level of byte grouping in an AXI-Stream interface.
- Data stream
  - The transport of data from one source to one destination.

TKEEP and TSTRB

- TKEEP
  - When TKEEP = 1, the associated byte must be transmitted to the destination
  - When TKEEP = 0, the associated byte is a null byte that can be removed from the stream
- TSTRB
  - When TKEEP = 1, TSTRB indicates whether the associated byte is a data byte or a position byte
  - When TSTRB = 1, the associated byte contains valid information and is a data byte
  - When TSTRB = 0, the associated byte does not contain valid information and is a position byte

Data width

D-03	D-07	D-0B	D-0F	D-13
D-02	D-06	D-0A	D-0E	D-12
D-01	D-05	D-09	D-0D	D-11
D-00	D-04	D-08	D-0C	D-10

Data width

Null	Null	D-07	D-0A	Null	D-0F
D-01	Null	D-06	D-09	Null	D-0E
Null	D-03	D-05	D-08	D-0C	Null
D-00	D-02	D-04	Null	D-0B	D-0D

Data width

D-03	D-07	Position	D-0F	D-13
Position	D-06	D-0A	Position	D-12
D-01	Position	D-09	D-0D	Position
D-00	D-04	Position	D-0C	D-10

TKEEP and TSTRB

TKEEP	TSTRB	Data Type	Description
HIGH	HIGH	Data byte	The associated byte contains valid information that must be transmitted between source and destination.
HIGH	LOW	Position byte	The associated byte indicates the relative position of the data bytes in a stream, but does not contain any relevant data values.
LOW	LOW	Null byte	The associated byte does not contain information and can be removed from the stream.
LOW	HIGH	Reserved	Must not be used.

TID and TDEST

- TID – Provides a stream identifier that **can** be used to differentiate between multiple streams of data that are being transferred across the same interface
- TDEST – Provides coarse routing information for the data stream
- Transfers that have the same TID and TDEST values are from the same stream

Interleaving and Ordering

Interleaving:

- Transfer interleaving is the process of interleaving transfers from different streams on a transfer-by-transfer basis
- TX can interleave streams at source and an interconnect **can** interleave streams from multiple TXs at a point of convergence
- RXs **can** be designed to accept any number of interleaved streams or a limited number of interleaved streams

Ordering:

- The AXI-Stream protocol **requires** that all transfers remain ordered
- The reordering of transfers **is not permitted**

Литература

- AMBA AXI-Stream Protocol Specification